Data Request: STADIOchoice Streaming Churn Prediction

Prepared for STADIOchoice, CAP182 Capstone SS1

This request covers the data needed to build a churn risk model for STADIOstream subscribers, as scoped in the accompanying problem statement. It is limited to STADIOstream households and does not include the satellite subscriber base or advertising logs, since the project targets streaming churn specifically and those sources do not speak directly to that problem at this stage. A consistent `household_id` across every table below is essential, since it is the key used to join all sources together.

Table 1: Subscription and Billing

Column name	Data type	Example	Additional information
household_id	String	HH-0043211	Unique identifier, must be consistent across every table in this request so records can be joined.
subscription_start_date	Date, yyyy-MM-dd	2023-01-15	Date the household first subscribed to STADIOstream.
plan_tier	String, categorical	Standard	Valid values limited to STADIOstream's current tier list, e.g. Basic, Standard, Premium.
monthly_price	Numeric, integer	14999	Price paid for the current tier, in cents.
billing_cycle_date	Date, yyyy-MM-dd	2025-05-03	Day of the month billing occurs.
payment_method	String, categorical	Debit order	Valid values: Credit card, Debit order, Mobile wallet.
payment_status	String, categorical	Failed	Status of the most recent billing attempt: Success, Failed, Pending.
payment_failures_last_90_days	Numeric, integer	2	Count of failed payment attempts in the trailing 90 days.
tenure_months	Numeric, integer	14	Months since subscription start.
upgrade_downgrade_flag	String, categorical	Downgrade	Tier change in the last billing cycle: Upgrade, Downgrade, None.
is_active	Boolean	TRUE	Whether the subscription is active as of the data extract date.
churn_date	Date, yyyy-MM-dd, nullable	2025-04-20	Cancellation date, blank if still active.

Table 2: Streaming Viewing Logs

Column name	Data type	Example	Additional information
household_id	String	HH-0043211	Joins to Table 1.
device_id	String	DEV-88213	Identifies the device used for the session.
event_timestamp	String, datetime, yyyy-MM-dd hh:mm:ss	2025-05-03 20:14:02	When the event occurred.
event_type	String, categorical	Play	Valid values: Play, Pause, Search, Scroll, Abandon.
title_id	String, nullable	TTL-2291	Joins to Table 5, blank for non-content events.
watch_duration_seconds	Numeric, integer, nullable	1830	Populated only for Play/Pause events.
device_type	String, categorical	Smart TV	Valid values: Mobile, Smart TV, Web, Tablet.

Table 3: Cancellation and Retention

Column name	Data type	Example	Additional information

household_id	String	HH-0043211	Joins to Table 1.
cancel_request_date	Date, yyyy-MM-dd	2025-04-18	Date the household requested cancellation.
cancel_reason_code	String, categorical	Price	Valid values: Price, Content, Technical issue, No reason given.
retention_offer_made	Boolean	TRUE	Whether a retention offer was extended.
retention_offer_type	String, categorical, nullable	Free month	Valid values: Discount, Free month, None.
retention_offer_accepted	Boolean, nullable	FALSE	Whether the household accepted the offer.
win_back_outcome	String, categorical, nullable	Churned	Valid values: Retained, Churned, Returned later.

Table 4: Customer Support

Column name	Data type	Example	Additional information
household_id	String	HH-0043211	Joins to Table 1.
contact_date	String, datetime	2025-04-10 09:32:00	When the contact occurred.
contact_channel	String, categorical	In-app chat	Valid values: Phone, Email, In-app chat.
reason_code	String, categorical	Billing dispute	Structured reason category assigned by the contact centre.
free_text_notes	String, text	"Customer says price increased without notice"	Raw agent notes or transcript summary, needed to mine for early warning patterns.
resolution_status	String, categorical	Resolved	Valid values: Resolved, Escalated, Unresolved.

Table 5: Content Catalogue

Column name	Data type	Example	Additional information
title_id	String	TTL-2291	Joins to Table 2.
title_name	String	The Great Rift	Display name of the title.
genre	String, categorical	Drama	Primary genre classification used by STADIOchoice.
is_sports_content	Boolean	FALSE	Flags sports content specifically, since sport tournament endings are a known churn trigger.
licensing_window_start	Date, yyyy-MM-dd	2024-01-01	Start of the licensing window.
licensing_window_end	Date, yyyy-MM-dd	2026-12-31	End of the licensing window.

Table 6: Search Queries

Column name	Data type	Example	Additional information
household_id	String	HH-0043211	Joins to Table 1.
search_timestamp	String, datetime	2025-04-09 21:03:11	When the search occurred.
search_term	String, text	The Great Rift season 2	Raw search text entered by the household.
result_count	Numeric, integer	0	Number of results returned, zero flags unmet demand.